

Extra - brief intro to measure-theoretic probability

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9:06 AM

Taken from Joel Tropp's CMS/ACM 117 lecture notes

Def: σ -algebra ("sigma" algebra)

X a domain, $\mathcal{F} \subseteq \mathcal{P}(X)$ is a σ -algebra on X
 $\uparrow$ power set

if 1) $\emptyset \in \mathcal{F}$, $X \in \mathcal{F}$ $\swarrow$ complement

2) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$

3) $A_i \in \mathcal{F}$ for $i \in \mathbb{N} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ and $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$

Borel σ -algebra on any separable metric space (eg., $\mathbb{R}^n$ w/ $\|\cdot\|_2$)

is the σ -algebra generated by open sets (or open intervals/balls)

Def measurable space is any $(X, \mathcal{F})$ w/ $\mathcal{F}$ a σ -algebra.

$A \in \mathcal{F}$ is called a "measurable" set

Def A measure μ is a function $\mu: \mathcal{F} \rightarrow [0, +\infty]$ s.t.

1) $\mu(\emptyset) = 0$

2) If A_i are disjoint measurable sets then $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$
"countable additivity"

Def. A measure space is a triple $(X, \mathcal{F}, \mu)$

Facts

• $A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$

• $\mu(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$ (A_i not necessarily disjoint)

• If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$ then $\mu(\bigcup_{i=1}^{\infty} A_i) = \lim_{i \rightarrow \infty} \mu(A_i)$

• If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$ and $\mu(A_{j_0}) < \infty$ for some j_0 , then $\mu(\bigcap_{i=1}^{\infty} A_i) = \lim_{i \rightarrow \infty} \mu(A_i)$

Def A probability measure is a measure μ st. $\mu(X) = 1$

"Def" On $X = \mathbb{R}$, the Lebesgue measure λ is the only Borel measure st. $\lambda((a, b]) = |b - a|$ and is translation invariant.

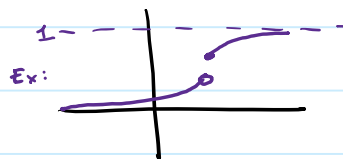
Think "uniform" density.

Def The (cumulative) distribution function "cdf" of a finite Borel measure μ on $X = \mathbb{R}$ is $F_{\mu}(a) := \mu((-\infty, a]) \quad \forall a \in \mathbb{R}$

Warning: no discussion or context given in my notes. This is only a cheat-sheet

CDF Facts

- $a \leq b \Rightarrow F_\mu(a) \leq F_\mu(b)$
- $\lim_{a \rightarrow -\infty} F_\mu(a) = 0, \quad \lim_{a \rightarrow +\infty} F_\mu(a) = \mu(\mathbb{R})$ } i.e. $= 1$ i.e. μ is a probability measure
- right continuous: $\lim_{a \downarrow x} F_\mu(a) = F_\mu(x)$
- $\mu((a, b]) = F(b) - F(a)$



Def $f: \mathbb{R} \rightarrow \mathbb{R}$ is a measurable function if $f^{-1}((t, \infty))$ is a Borel-set

Def The Lebesgue integral of a positive, measurable function, wrt Borel measure μ , is

$$\int_{\mathbb{R}} f(x) \mu(dx) := \lim_{R \rightarrow \infty} \underbrace{\int_0^R \mu(\{x: f(x) > t\}) dt}_{\text{Riemann integral}}$$

↑
need not be Lebesgue measure

There are other ways to define

Def A measurable finite-valued function f is integrable if $\int |f(x)| \mu(dx) < \infty$
then define

aka $f \in L_1(\mu)$

$$\int f(x) \mu(dx) = \int f_+(x) \mu(dx) - \int f_-(x) \mu(dx)$$

Facts

- Integration is a linear operator

[Skip lecture 5, 6 in Trapp]

$$\Omega = X \quad \mathbb{P} = \mu$$

Def A probability space is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$

Ω is sample space, $\omega \in \Omega$ a sample point or "outcomes"

$\mathcal{F}$ are events

$$\mathbb{P}(\Omega) = 1$$

Def A real random variable is a measurable function $X: \Omega \rightarrow \mathbb{R}$

"r.v."

i.e. an encoding. If $\Omega = \{H, T\}$ (Heads, Tails, for a single coin flip)

then $X(\omega) = \begin{cases} 1 & \text{if } \omega = H \\ -1 & \text{if } \omega = T \end{cases}$ is an example of a random variable

Def The law of a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ is the Borel measure μ_X

defined by $\mu_X(B) := \mathbb{P}[X^{-1}(B)] = \mathbb{P}[X \in B] \quad \forall \text{ Borel } B \text{ on } \mathbb{R}$

aka distribution of X

$$= \int_{\mathbb{R}} \mathbb{1}_B(x) \mu_X(dx) = \int_{\mathbb{R}} \mathbb{1}_B d\mu_X = \int_B d\mu_X$$

Def The distribution function of a (real) random variable is

$$F_X(a) := P[X \leq a] = \mu_X(-\infty, a] \quad \forall a \in \mathbb{R}$$

aka cdf for cumulative distribution function

Has same monotonicity properties, etc., as the generic distribution function
(for any old measure μ) that we already saw.

Absolutely continuous r.v. / distributions " $\mu_X \ll \lambda$ "

means the law μ_X has a density f_X w.r.t. the Lebesgue measure

$$\text{so that } \mu_X(B) = \int_B d\mu_X = \int_B f_X(x) \lambda(dx)$$

we can
always do
this

only if
abs. cts.

$$\text{i.e. } P[X \in B] = \int_B \underbrace{f_X(x)}_{\text{"pdf"}} dx$$

Same thing as what people have in mind if they
just say a "continuous" r.v.

Ex: exponential, Laplace,
Normal, Gamma, Beta

⚠ $\mu_X(B) = \int_B d\mu_X$ is always true,

since $\int$ is our fancy Lebesgue integral which
allows for nontraditional integration (like
dirac delta functions)

Def A pair of r.v. (X, Y) on $(\Omega, \mathcal{F}, P)$ just

means both X and Y are r.v.

Their joint law is $\mu_{XY}(B) = P[(X, Y) \in B] \quad \forall$ Borel sets $B \subseteq \mathbb{R}^2$

The marginal laws are $\mu_X(B) = P[X \in B] \quad \forall$ Borel sets $B \subseteq \mathbb{R}$
and likewise for μ_Y

We say X and Y are independent if $\mu_{XY} = \mu_X \times \mu_Y$

$$\text{aka } P[(X, Y) \in A \times B] = P[X \in A] \cdot P[Y \in B]$$

Def The expectation of a r.v. X is

$$E[X] = P[X] = \int_{\Omega} X(\omega) P(d\omega) = \int_{\Omega} X dP \quad \dots \text{and it's linear!}$$